WEIR - Assignment 2 - Document Retrieval

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***Abstract*—This assignment builds on the PA1 crawl by extracting meaningful content from HTML pages, segmenting it into coherent text segments, encoding them with a sentence embedding model, and storing them in a pgvector-enabled PostgreSQL database for semantic retrieval.**

We implement parsing tailored to 24ur.com articles, apply sentence-aware chunking, index embeddings with HNSW, and provide a query and reranking pipeline based on vector similarity and a cross-encoder reranker.

I. INTRODUCTION

This report describes the design and implementation of a parsing and semantic embedding pipeline that processes the HTML pages collected by the preferential web crawler developed in the previous assignment. The goal is to extract clean article text from the crawled pages and represent it in a form that supports efficient domain-specific information retrieval.

The pipeline consists of two main stages: preprocessing and retrieval. During preprocessing, crawled HTML pages from 24ur.com are parsed, and the extracted text is divided into sentence-based segments. The resulting segments are then embedded using a transformer-based sentence embedding model and stored in a vector database.

During retrieval, a user query is embedded using the same embedding model and used to retrieve the most similar segments from the database. Query–segment pairs are subsequently evaluated by a reranking model, which refines the initial ranking based on semantic relevance. The final output of the pipeline is a ranked list of text segments that are most semantically relevant to the query.

II. METHODOLOGY

A. Website filtering and data source

We reuse the PA1 crawl database from the previous assignment and restrict processing to HTML pages with non-empty `html_content`. In our pipeline this filtering is performed in a database query that fetches page IDs and HTML bodies.

B. HTML parsing with XPath and regural expressions

HTML parsing is implemented with `lxml`. We extract the article title by first reading the OpenGraph metadata (`//meta[@property='og:title']/@content`); if missing, we fall back to `//title`. A regex post-process removes the site suffix "`| 24ur.com`" and normalizes whitespace. The main article body is extracted from the `#article-body` node using XPath, and non-content elements (`script`, `style`, `noscript`, `img`, `figure`, `svg`, `picture`) are removed to keep only the textual content.

We additionally capture the lead summary by locating the main article picture and the adjacent summary paragraph via XPath, then prepend it to the article body. A regex is used for lightweight text normalization (whitespace collapsing) and title cleanup.

This hybrid approach keeps extraction robust while remaining lightweight and transparent.

C. Chunking

After parsing the crawled web pages, the extracted text is divided into smaller sentence-based segments. Sentence boundaries are detected using the NLTK sentence tokenizer configured for the Slovene language. Segments are constructed incrementally by appending consecutive sentences until a predefined maximum word limit is reached. Once the limit is exceeded, a new segment is created. This approach preserves local semantic context while ensuring that segment sizes remain suitable for transformer-based embedding models.

D. Segment Embedding

Document segments were transformed into dense vector representations using transformer-based sentence embedding models. Each segment was embedded independently, and the resulting vectors were stored in dedicated database tables grouped by embedding dimensionality. We extend the PA1 database with a `cleaned_content` column and three segment tables for different embedding sizes: `page_segment_vec384`, `page_segment_vec768`, and `page_segment_vec1024`. Each segment row stores the page ID, the model ID, the text segment, and the vector embedding. A separate `model` table records the embedding model name, enabling multiple models to coexist.

To ensure consistent similarity computation during retrieval, embeddings generated by all evaluated models were normalized to unit length prior to storage, despite the embedding models already producing nearly unit-length vectors.

The embedding computation included both multilingual and Slovenian-specialized embedding models. Table I summarizes the embedding models used in the experiments together with their output vector dimensionalities.

TABLE I: Embedding models used for vectorization

Model	**Vec. Length**
`sentence-transformers/all-MiniLM-L6-v2`	384
`sentence-transformers/all-mpnet-base-v2`	768
`EMBEDDIA/sloberta`	768
`cjvt/crosloengual-bert-si-nli`	768
`BAAI/bge-m3`	1024

To support efficient similarity-based retrieval, vector indexes were created for each embedding table using the `pgvector` PostgreSQL extension. Separate HNSW indexes were generated for cosine similarity and Euclidean (L2) distance metrics, enabling experimentation with different retrieval strategies. Additionally, standard indexes were created for embedding model identifiers to enable efficient filtering by model. Since `pgvector` does not provide indexing support for L1 distance, L1 retrieval is performed without vector indexing.

E. Segment Retrieval and Reranking

During retrieval, the user query is embedded using the same embedding model as the stored segment embeddings. The generated query vector is then used to query the vector database and retrieve the top 100 most similar segments according to the selected distance metric.

The retrieved segments are subsequently reranked using a cross-encoder model. For reranking, query–segment text pairs are constructed and processed jointly by the reranker model to produce relevance scores. The final retrieval output consists of the 10 highest-ranked segments according to the reranker scores.

Table II summarizes the reranking models used in the experiments.

TABLE II: Reranking model names

Reranker Model
`cross-encoder/ms-marco-MiniLM-L6-v2`
`cross-encoder/mmarco-mMiniLMv2-L12-H384-v1`
`BAAI/bge-reranker-v2-m3`

III. Embedding Analysis

To evaluate the retrieval quality of our embedding models, we employ the Normalized Discounted Cumulative Gain metric at k retrieved segments (NDCG@k). This metric is well suited for retrieval pipelines, as it measures not only the number of relevant segments among the top k retrieved results, but also whether the most relevant segments appear near the top of the ranking. For evaluation, we compute the mean NDCG@k across multiple queries, providing a more representative measure of average retrieval performance.

A. Baseline Embedding Model Retrieval Performance

First, we evaluated the quality of the embedding models by retrieving N closest segment embeddings to each of the query embeddings according to the cosine distance metric. This provides a baseline performance measure for selecting the best-performing embedding model without introducing bias from reranking models.

Queries used when computing mean NDCG@k scores were obtained by randomly sampling pages from the database and taking the first segment of each page. Only the first 128 characters of the chosen segments were used to extract a portion of the title. After we retrieved N closest segments to each query (for each model), ground truth relevancy score was calculated with different strategies: BM25 score, higher score for same page segments, and comparing to a reranked order using one of the reranking models. We present measurement for each strategy separately and don't combine them because the final performance measurements would be greatly influenced by how all three strategies were weighted. More about the reasons for choosing such strategies is told in the Section V.

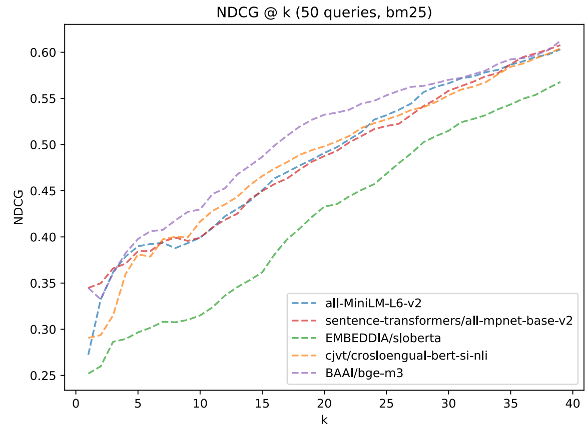

Fig. 1: Comparison of embedding models using the mean NDCG@k metric with the BM25 strategy.

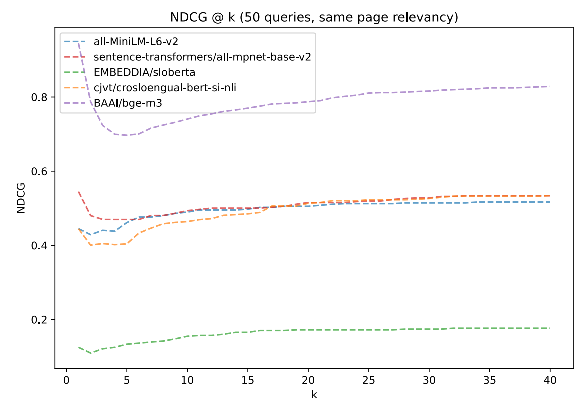

Fig. 2: Comparison of embedding models using the mean NDCG@k metric with the same page strategy.

For the BM25 strategy, we used the "Lucene" version of the BM25 algorithm and computed scores for each result. The top 10 highest scoring results kept their scores, while others were set to 0 to emphasize the difference between relevant and irrelevant results. Measurements using this strategy are shown in Figure 1.

The second strategy focused on giving higher scores to segments that were taken from the same page as the query since it makes sense that segments on the same page talk about the same thing. Therefore segments on the same page as the query are given a score 1, otherwise 0. Measurements using this strategy are shown in Figure 2.

The third strategy simply compares orderings of each

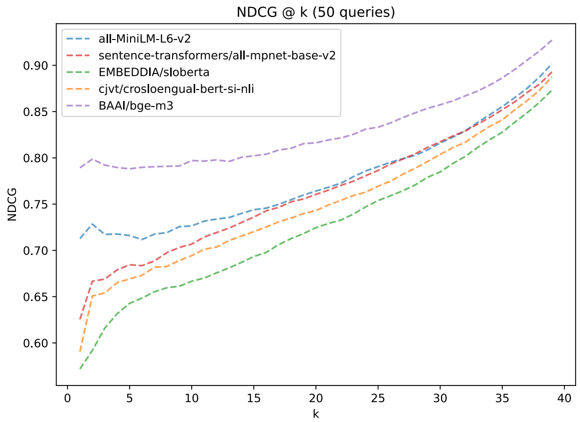

Fig. 3: Comparison of embedding models using the mean NDCG@k metric with the comparison of retrieved and reranked orderings strategy.

embedding model with their reranked orderings using the `ms-marco-MiniLM-L6-v2` model. An embedding model that produced an ordering closer to its reranked ordering is considered better. Measurements using this strategy are shown in Figure 2.

Our final evaluation strategy used an LLM to semantically assess the relevance of retrieved segments to a query. Each segment was assigned a relevance score from 0 (irrelevant) to 4 (highly relevant and directly related to the query). To emphasize highly relevant results in the ranking metric, the scores were transformed using an exponential function before computing NDCG@k.

For evaluation, `gpt-4.1-mini` was used to score all 50 retrieved segments for each query in a single pass. NDCG@k was then computed for values of k ranging from 2 to 50. Figure 4 presents the mean retrieval performance of the evaluated embedding models using cosine similarity over 20 queries.

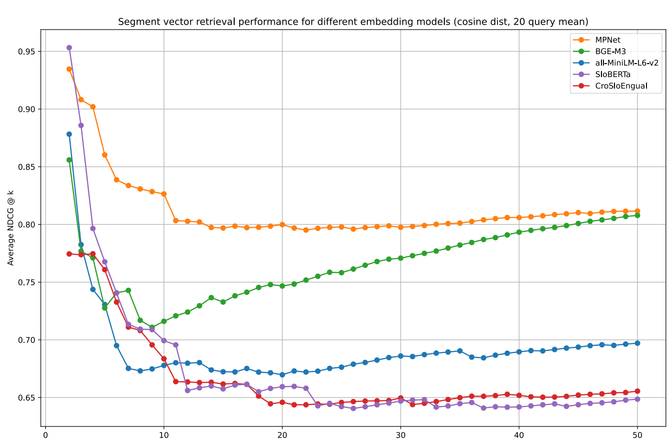

Fig. 4: LLM-based NDCG@k evaluation of embedding models using cosine similarity.

For the final retrieval pipeline, we selected the `bge-m3` embedding model due to its consistently strong performance across all evaluations.

B. Distance Metric Performance

We evaluated the impact of different distance metrics for measuring embedding similarity. Cosine, L1, and L2 distances were compared using the `bge-m3` embedding model. The evaluation followed the same LLM-based scoring procedure described previously, with NDCG@k averaged over 50 queries.

Figure 5 shows the relationship between the evaluated metrics. Since all embeddings were normalized, cosine similarity and L2 distance produced identical retrieval rankings and therefore nearly identical NDCG@k scores. In contrast, the L1 distance consistently underperformed, likely due to the curse of dimensionality in high-dimensional embedding spaces and the lost magnitude in embeddings due to normalization. Based on these results, cosine similarity was selected for the final retrieval pipeline.

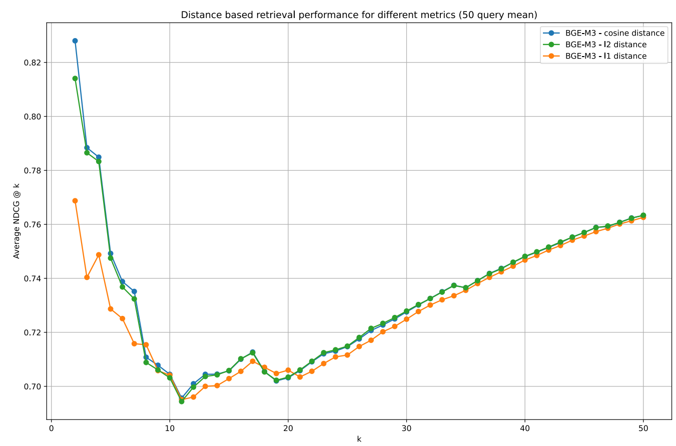

Fig. 5: Comparison of distance metrics using LLM-based NDCG@k evaluation.

C. Reranker Performance

Finally, we evaluated the performance of all embedding model and reranker combinations by computing the mean NDCG@k over 20 queries. Ground-truth relevance scores were obtained using the same LLM-based evaluation procedure described previously. For each query, the top 200 segments were first retrieved using cosine similarity on the embedding vectors, after which the reranker models reordered the results and returned the top 50 segments.

Figure 6 presents the performance of all evaluated combinations. The `mmarco-mMiniLMv2-L12-H384-v1` reranker consistently provided the largest performance improvements across embedding models. In contrast, `bge-reranker-v2-m3` achieved lower performance despite being the largest evaluated reranker model. We attribute this to weaker Slovene language representation during training, despite the model being advertised as multilingual.

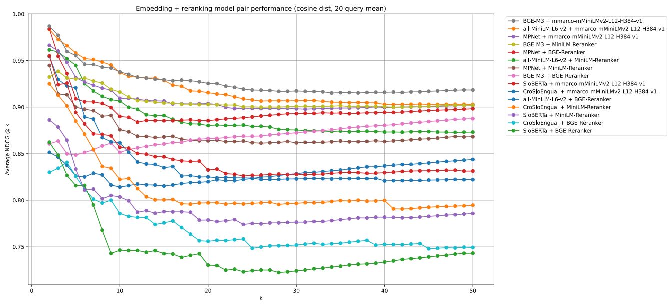

Fig. 6: LLM-based NDCG@k evaluation of embedding model and reranker combinations.

The best overall performance was achieved by combining the `bge-m3` embedding model with the `mmarco-mMiniLMv2-L12-H384-v1` cross-encoder reranker. Due to their consistently strong retrieval performance, this combination was selected for the final retrieval pipeline.

Additionally, we observed that the NDCG@k scores plateaued for nearly all evaluated models after $k = 10$. This suggests that retrieving additional segments beyond the top 10 provides little improvement in retrieval quality. Based on this observation, we selected 10 as the final number of retrieved segments in the final pipeline.

D. Embedding Visualization

To evaluate how effectively the models group embedded segments, we visualized embeddings using UMAP. The embeddings were first reduced with SVD, retaining 40% of the variance, and then projected into a two-dimensional space using UMAP. Each point in the scatter plot represents a single segment embedding. Points of the same colors represent results of the same query, where the query point is marked as "X".

IV. Results

The final selected parameters of our pipeline are noted in Table III.

The results indicate that the model `bge-m3` [1] consistently outperforms the other evaluated models, although much less convincingly according to the BM25 strategy. Cosine similarity was used for all similarity-based retrieval experiments. Among all models, `sloberta` was the worst. Surprisingly, `all-MiniLM` was as good or better that all 768-dimensional embedding models while having twice less embedding dimensions.

Unsurprisingly, query results of the `bge-m3` model are also better grouped than other models. The worst grouping comes from the `sloberta` model, where many queries are always put on the same island, indicating the model is having a hard time separating them.

- NDCG @ k metric for final params - examples of parsed html - examples of segments

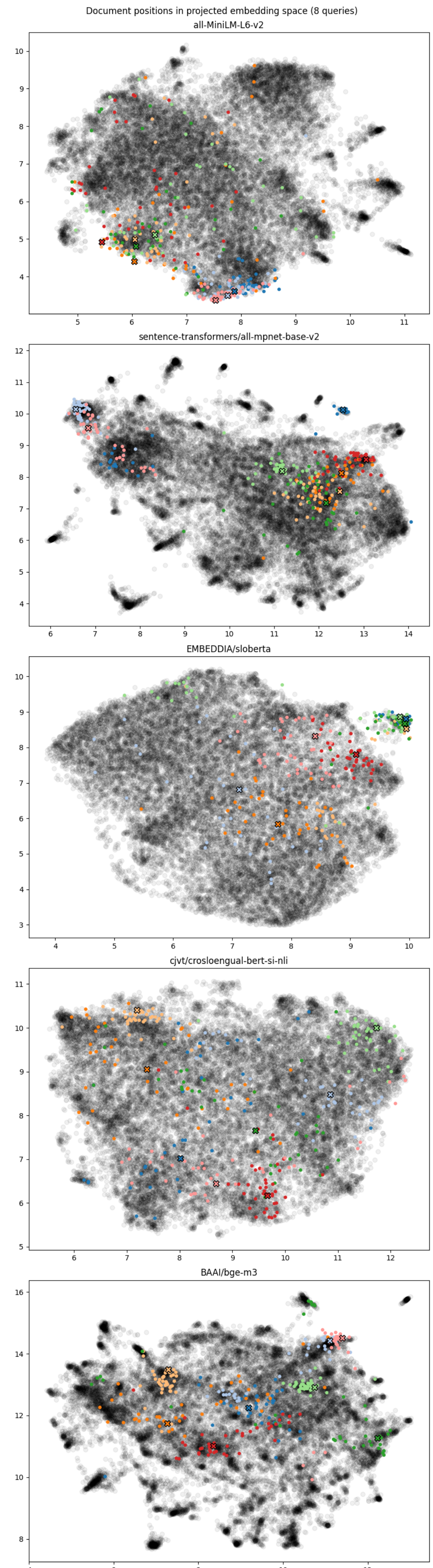

Fig. 7: UMAP visualization of segment embedding clusters.

TABLE III: Final pipeline parameters

Parameter	Chosen Value
`Embedding Model`	BAAI/bge-m3
`Reranking Model`	mmarco-mMiniLMv2-L12-H384-v1
`Segment Max Length`	128 words
`Retrieval Dist. Metric`	cosine
`Similarity Return n`	100
`Reranking Return k`	10

A. *Retrieval Examples*

- examples of successful queries - examples of unsuccessful queries

V. DISSCUSSION

For measuring retrieval quality of embedding and reranking models we would ideally have a manually annotated dataset of queries, their results, and relevance scores. Since we couldn't afford to make one, we opt to employ diferent algorithms and strategies to calculate ground truth relevance scores and measured models on each one. We think that the least reliable strategy was the BM25 one since embedding models are expected to encode meanings of the tet other that just simple term frequencies.

Experiments clearly show reranking significantly improves the relevancy of top results, but due to the harder interpretability of the NDCG@k metric concrete relative improvements between models are harder to determine.

UMAP visualizations helped us somewhat validate our measurements and we did keep in mind that the visualization inaccurately represents the real embedding space. We hoped however that if a model was able to extract many useful features and group segments around them, we would see more groups in the UMAP visualization as well.

- what can be done better, what we could expand on - TODO EXPAND LATER The parser is tailored to 24ur.com and relies on stable HTML structure. Pages that deviate from the `#article-body` template or heavily script-rendered content can yield empty or partial text. Chunking relies on sentence tokenization quality and may produce uneven segment sizes. Reranking improves accuracy but adds latency, which may be problematic for interactive settings.

VI. CONCLUSION

We implemented an end-to-end extraction and retrieval pipeline for the PA1 crawl of 24ur.com. The system parses raw HTML, chunks text using sentence boundaries, stores embeddings in a pgvector database with HNSW indexing, and retrieves and reranks results for user queries. The approach is modular, supports multiple embedding models, and can be easily configured using an `.env` file.

REFERENCES

[1] J. Chen, S. Xiao, P. Zhang, K. Luo, D. Lian, and Z. Liu, "Bge m3-embedding: Multi-lingual, multi-functionality, multi-granularity text embeddings through self-knowledge distillation," *arXiv preprint arXiv:2402.03216*, vol. 4, no. 5, 2024.

APPENDIX

Rank	Retrieved Text	Distance
1	Vladimir putin.	0.0432
2	vladimir putin.	0.0432
3	Vladimir Putin.	0.0432
4	putin.	0.1489
5	Putin.	0.1489
6	vladimi putin.	0.1500
7	vladmir putin.	0.1539
8	Vladimir.	0.2432
9	vladimir.	0.2432
10	Litvinenko.	0.3473

TABLE IV: Top 10 retrieval results using the `all-MiniLM` model before reranking for query="Vladimir Putin".

Rank	Reranked Text	Cross Score	Distance
1	Vladimir putin.	7.190	0.0432
2	vladimir putin.	7.190	0.0432
3	Vladimir Putin.	7.190	0.0432
4	Novoizvoljen ruski predsednik. Vladimir Putin je po osmih letih vladanja krmilo države predal 13 let mlajšemu političnemu varovancu Dmitirju Medvedjevu. Ruski p...	6.703	0.4193
5	Putin je optimističen: Najhuje je mimo. Ruski predsednik Vladimir Putin je na tradicionalnem televizijskem maratonu odgovarjal na vprašanja državljanov. Vprašan...	6.672	0.4288
6	Ruski predsednik Vladimir Putin in njegov ukrajinski kolega Volodimir Zelenski sta sicer med prvim telefonskim pogovorom v začetku meseca govorila o možnosti za...	6.504	0.4321
7	1 7 Vladimir Putin in venezuelski predsednik Nicolas Maduro APVladimir Putin in kitajski predsednik Ši Džinping ProfimediaVladimir Putin in brazilski predsednik...	6.268	0.4156
8	Putin: Nobena država nima česa takšnega, raketa je neranljiva za zračno obrambo. ZDA: Smo povsem pripravljeni. Vladimir Putin je v letnem nagovoru poslancem spo...	6.061	0.4377
9	vladimi putin.	5.999	0.1500
10	Zakaj je Putin med obiskom v Moskvi Stevu Witkoffu predal red Lenina?. Ruski predsednik Vladimir Putin je med obiskom v Moskvi ameriškemu posebnemu odposlancu p...	5.731	0.4451

TABLE V: Top 10 retrieval results using the `all-MiniLM` model after reranking with the `ms-marco-MiniLM` model for query="Vladimir Putin".

Rank	Reranked Text	Cross Score	Distance
1	Vladimir Putin.	1.0000	0.0432
2	Vladimir putin.	0.9999	0.0432
3	vladimir putin.	0.9999	0.0432
4	vladmir putin.	0.9999	0.1539
5	vladimi putin.	0.9998	0.1500
6	Putin.	0.9996	0.1489
7	Vladimir.	0.9987	0.2432
8	vladimir.	0.9984	0.2432
9	Vladislav.	0.9982	0.4304
10	vladimir pustin.	0.9979	0.3915

TABLE VI: Top 10 retrieval results using the `all-MiniLM` model after reranking with the `bge-reranker` model for query="Vladimir Putin".

Rank	Retrieved Text	Distance
1	Rusi demonstrirali 'povratni jedrski napad': uporabili balistične rakete. Ruski predsednik Vladimir Putin je prek digitalne platforme spremljal vojaške vaje rus...	0.2464
2	Prigožinov upor je predstavljal enega največjih izzivov za dolgoletno vladavino Vladimirja Putina. Vodja Wagnerjeve skupine je imel pomembno vlogo pri invaziji ...	0.2496
3	Pele javno poziva Putina k zaustavitvi vojne. Brazilska nogometna legenda Pele je ruskega predsednika Vladimirja Putina javno pozval, naj konča vojno v Ukrajini...	0.2525
4	Putin se zaradi oslabljene imunosti verjetno boji okužbe, so špekulirali. Pred kratkim so se pojavile tudi govorice, da Putina nenehno spremlja zdravnik, ki je ...	0.2540
5	Zdi se, da Američani niso pripravljeni iti predaleč proti Rusiji, Evropejci pa še naprej podpirajo Ukrajino, zato smo v zastoju. "Kakšna je evropska pogajalska ...	0.2602
6	Rusija je obtožbe zavrnila in poudarila, da je odločitev Ovseja znak naraščajočih napetosti med Rusijo in Zahodom. Putin je tako izjavil, da za bojkotom zahodni...	0.2625
7	EU je v obrazložitvi, zakaj je 69-letni milijarder pristal na seznamu sankcioniranih, navedla, da je Usmanov "prokremeljski oligarh s posebno tesnimi povezavami...	0.2636
8	"Putin tukaj ustvarja določeno vzdušje, katerega cilj je spodkopavanje zaupanja javnosti v vlado, v ključne strukture, kot so oborožene sile in policija, ter us...	0.2642
9	Ukrajinska vlada je vztrajala pri prisotnosti poveljnika upornikov v Donecku Aleksandra Zaharčenka in voditelja separatistov v Lugansku Igorja Plonickija. O dana...	0.2643
10	Zelenski Rusom namenil sporočilo: Putin vas uničuje. Iz več ukrajinskih regij poročajo o raketnih napadih. V Kijevu je odjeknilo deset eksplozij. Za približno p...	0.2647

TABLE VII: Top 10 retrieval results using the `all-MiniLM` model before reranking for query="Zelenski Zahod poziva k odzivu, Putin: Rakete so pripravljene za uporabo.".

Rank	Reranked Text	Cross Score	Distance
1	Zelenski Zahod poziva k odzivu, Putin: Rakete so pripravljene za uporabo. Ruski predsednik Vladimir Putin je sporočil, da bo nadaljeval s testiranjem balistični...	9.227	0.2823
2	Zelenski Rusom namenil sporočilo: Putin vas uničuje. Iz več ukrajinskih regij poročajo o raketnih napadih. V Kijevu je odjeknilo deset eksplozij. Za približno p...	6.814	0.2647
3	Rusi demonstrirali 'povratni jedrski napad': uporabili balistične rakete. Ruski predsednik Vladimir Putin je prek digitalne platforme spremljal vojaške vaje rus...	6.441	0.2464
4	Dodal je, da so se z ZDA pripravljeni srečati na ravni voditeljev držav, da bi obravnavali občutljiva vprašanja. "O zadevah, kot so ozemeljska vprašanja, se je ...	5.991	0.2833
5	Na 1000. dan vojne v Ukrajini Putin podpisal odlok za širšo rabo jedrskega orožja. Ruski predsednik Vladimir Putin je podpisal odlok, ki Moskvi omogoča uporabo ...	5.791	0.2684
6	"Predsednik Putin je v nagovoru narodu dejal, da je regionalni konflikt v Ukrajini, kot ga je poimenoval, po napadih na rusko regijo Brjansk in Kursk z ameriški...	5.408	0.2744
7	"Ne vidimo nobenih znakov, da se Rusija pripravlja na uporabo jedrskega orožja," je po Putinovih pripombah dejal ameriški državni sekretar Antony Blinken. Za ogl...	5.368	0.2784
8	Prav tako naj bi šlo šele za prvi val raketnih napadov. Guverner ukrajinske regije Mikolajev Vitalij Kim je že predhodno opozoril na možnost raketnih napadov. Uk...	5.177	0.2764
9	Prigožinov upor je predstavljal enega največjih izzivov za dolgoletno vladavino Vladimirja Putina. Vodja Wagnerjeve skupine je imel pomembno vlogo pri invaziji ...	5.081	0.2496
10	Generalni sekretar Nata Jens Stoltenberg je medtem zaradi Putinove odločitve izrazil zaskrbljenost. Sile za odvračanje so enote, katerih namen je odvrniti napad ...	4.513	0.2818

TABLE VIII: Top 10 retrieval results using the `all-MiniLM` model after reranking with the `ms-marco-MiniLM` model for query="Zelenski Zahod poziva k odzivu, Putin: Rakete so pripravljene za uporabo.".

Rank	Reranked Text	Cross Score	Distance
1	Zelenski Zahod poziva k odzivu, Putin: Rakete so pripravljene za uporabo. Ruski predsednik Vladimir Putin je sporočil, da bo nadaljeval s testiranjem balistični...	0.9993	0.2823
2	Prav tako naj bi šlo šele za prvi val raketnih napadov. Guverner ukrajinske regije Mikolajev Vitalij Kim je že predhodno opozoril na možnost raketnih napadov. Uk...	0.9129	0.2764
3	Dodal je, da so se z ZDA pripravljeni srečati na ravni voditeljev držav, da bi obravnavali občutljiva vprašanja. "O zadevah, kot so ozemeljska vprašanja, se je ...	0.6005	0.2833
4	Rusi demonstrirali 'povratni jedrski napad': uporabili balistične rakete. Ruski predsednik Vladimir Putin je prek digitalne platforme spremljal vojaške vaje rus...	0.5256	0.2464
5	Na 1000. dan vojne v Ukrajini Putin podpisal odlok za širšo rabo jedrskega orožja. Ruski predsednik Vladimir Putin je podpisal odlok, ki Moskvi omogoča uporabo ...	0.4899	0.2684
6	"Predsednik Putin je v nagovoru narodu dejal, da je regionalni konflikt v Ukrajini, kot ga je poimenoval, po napadih na rusko regijo Brjansk in Kursk z ameriški...	0.3710	0.2744
7	Zelenski Rusom namenil sporočilo: Putin vas uničuje. Iz več ukrajinskih regij poročajo o raketnih napadih. V Kijevu je odjeknilo deset eksplozij. Za približno p...	0.2555	0.2647
8	Ruski predsednik Vladimir Putin je v ponedeljek obljubil "primeren odgovor" na napad, katerega motivacijo je predstavil kot pretežno politično. "Naša glavna nal...	0.2171	0.2744
9	Putin: Rusija bo jedrsko orožje uporabila le, če bo napadena. Ruski predsednik Vladimir Putin je v nagovoru na letnem zasedanju kremeljskega sveta za človekove ...	0.1816	0.2832
10	Generalni sekretar Nata Jens Stoltenberg je medtem zaradi Putinove odločitve izrazil zaskrbljenost. Sile za odvračanje so enote, katerih namen je odvrniti napad ...	0.1690	0.2818

TABLE IX: Top 10 retrieval results using the `all-MiniLM` model after reranking with the `bge-reranker` model for query="Zelenski Zahod poziva k odzivu, Putin: Rakete so pripravljene za uporabo.".